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Dynamic analysis of the NACA0015 airfoil : A numerical approach

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Abstract

Accurate turbulence modelling is critical to the effective solution of aerodynamic problems, particularly in the prediction of dynamic stall behaviour. This study focuses on the aerodynamic performance of the NACA0015 airfoil at different angles of attack with the aim of understanding and analysing dynamic stall phenomena. First, a numerical simulation of the turbulent flow around a stationary airfoil is performed to determine the static stall angle, then a dynamic mesh analysis is performed to study the airfoil under oscillating conditions, and finally the study is extended to a vertical axis Darrieus wind turbine using OpenFOAM based on the finite volume method, with particular emphasis on the selection of an appropriate turbulence model to ensure the highest accuracy in capturing the flow characteristics.

Introduction

The study of aerodynamic performance is crucial in various engineering applications, particularly in the design and optimization of airfoils. This document presents a detailed numerical analysis of the NACA0015 airfoil, focusing on its behavior under varying angles of attack. Accurate turbulence modeling is essential for predicting dynamic stall phenomena, which can significantly impact the performance and efficiency of aerodynamic systems.

Governing equations

To numerically study the flow around the NACA airfoil, we will use the Reynolds Average Navier Stokes (RANS) equations. These equations are written as follows.

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho U_i)}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial(\rho U_i)}{\partial t} + \frac{\partial(\rho U_i U_j)}{\partial x_i} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_i}\left[\mu \frac{\partial U_j}{\partial x_i} + \frac{\partial U_i}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial U_i}{\partial x_i}\right] + \frac{\partial(-\rho u_i u_j)}{\partial x_i} \quad (2)$$

Where :

U_i, U_j: instantaneousvelocity . u_i, u_j: Velocity fluctuationdue to turbulence.

ρ : Density of air. P : Pressure

δ_{ij} :Kronecker Delta

The equations used to model turbulence are the k-ω SST, which is actually a "hybridised" model that combines the strengths and superior behaviour of the k-ε and k-ω models.

$$\frac{\partial k}{\partial t} + U_i \frac{\partial k}{\partial x_i} = P_k - \beta k \omega + \frac{\partial}{\partial x_i}\left[(\nu + \sigma_k \nu_t)\frac{\partial k}{\partial x_i}\right] \quad (3)$$

$$\frac{\partial \omega}{\partial t} + U_i \frac{\partial \omega}{\partial x_i} = \alpha S^2 - \beta \omega^2 + \frac{\partial}{\partial x_i}\left[(\nu + \sigma_\omega \nu_t)\frac{\partial \omega}{\partial x_i}\right] + 2(1 - F_1)\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial \omega}{\partial x_j}\frac{\partial k}{\partial x_j} \quad (4)$$

Numerical results

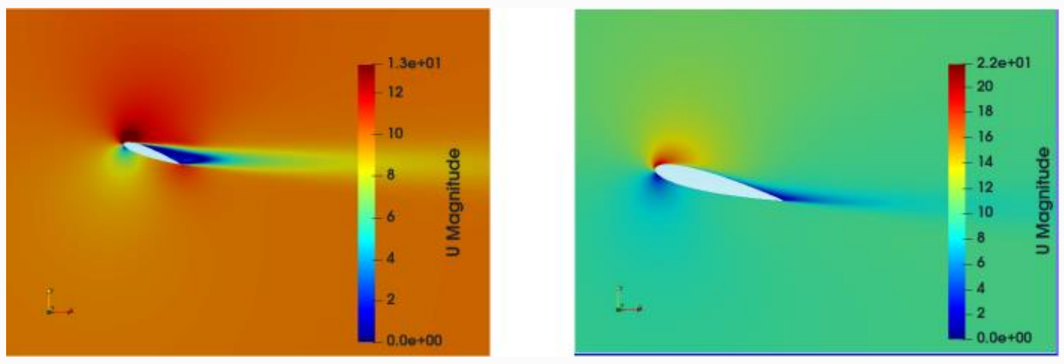

Figure 1 :velocity contour for Re = 6.6 10^5 for an angle of attack of 22° and 14°.

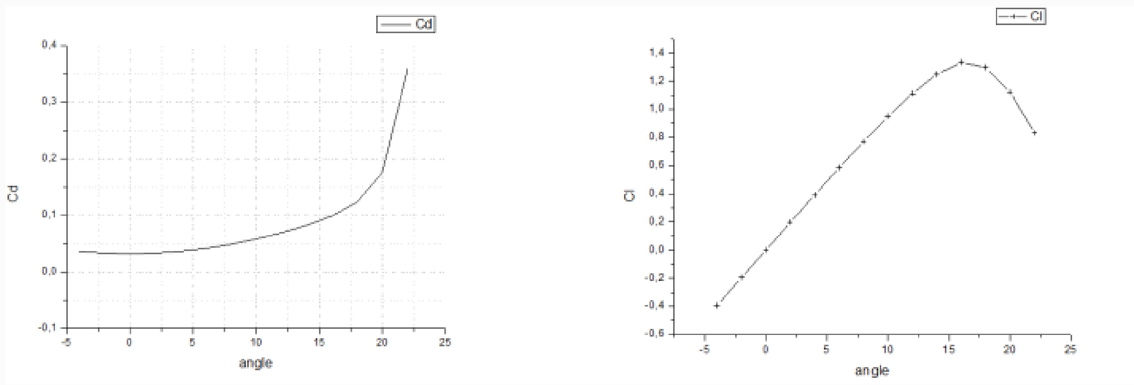

Figure 2: Drag and lift coefficient variation with angle of attack

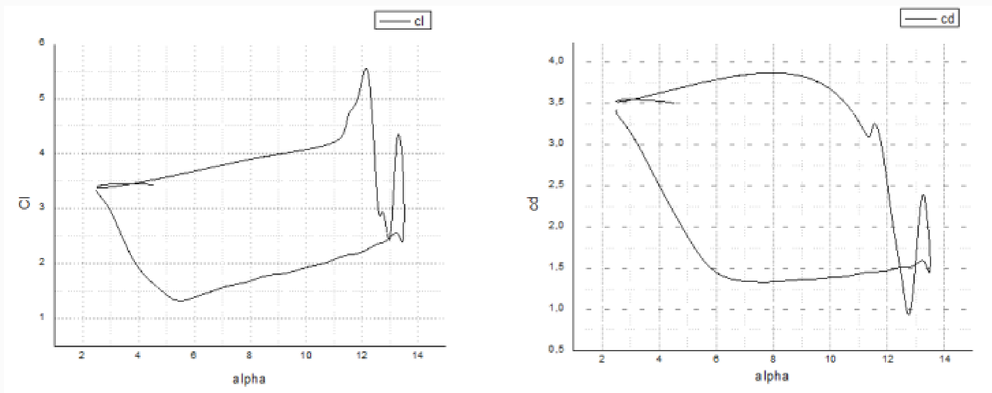

Figure 3: Variation of drag and lift coefficient as a function of angle of attack for the dynamic mesh case

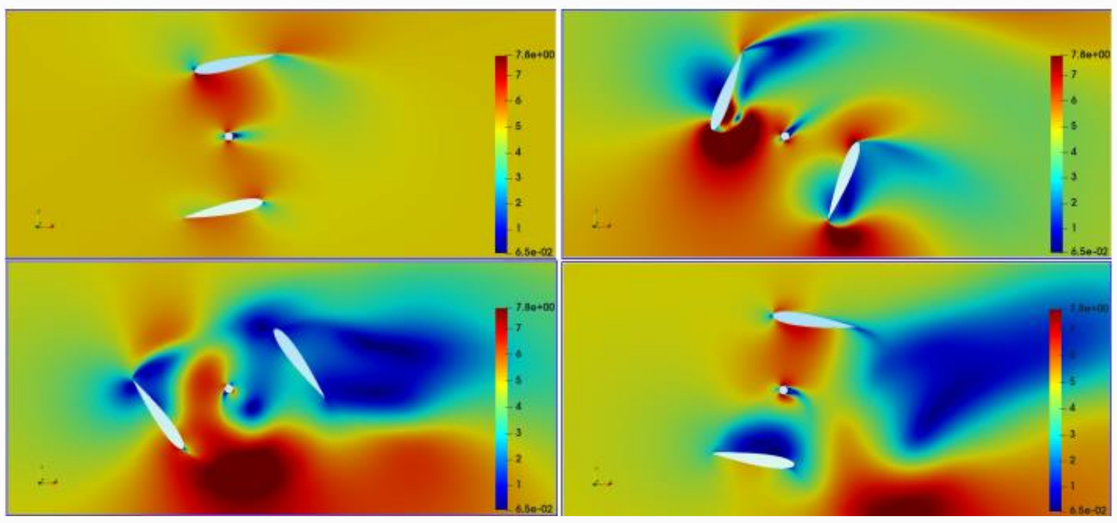

Figure 4: Visualization of the simulation showing the variation in speed around the wind turbine

Conclusion

The results obtained from these simulations show good agreement with existing literature, both experimental and numerical, further validating the reliability of the chosen methods and models.

Références

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